

**TED (15)-3181**  
**(Revision—2015)**

**Reg. No. ....**

**Signature .....**

**THIRD SEMESTER DIPLOMA EXAMINATION IN ARCHITECTURE**  
**MODEL QUESTION PAPER**  
**CLIMATOLOGY**

**Time – 3 Hrs]**

**[Max.Marks-100]**

**PART- A**  
**(Maximum Marks: - 10)**

I. Answer the following questions in or two sentences. Each question carries 2 marks

1. Outline the influence of topography on air movement.
2. Define inter tropical convergence zone.
3. Write on bioclimatic chart.
4. Define evaporative cooling.
5. Traditional shelters in tropical upland climate is circular. Give reason.

**PART B**  
**(Maximum Marks: - 30)**

II Answer any five questions from the following. Each question carries 6 marks.

1. Describe the data to be collected related to site for preparing a design.
2. Define the following. 1. Trade winds 2. Westerlies. 3. Polar winds.
3. Discuss on ET scale and CET scale.
4. Explain time-lag and decrement factor.
5. Describe envelope design.
6. Illustrate with sketch working of domestic refrigerator.
7. Explain the purpose of cross ventilation in providing physiological cooling in warm humid climate. (5 x 6 marks)

**PART- C**  
**(Maximum marks :- 60)**

Answer one full question from each unit. Each full question carries 15 marks

**UNIT – I**

- III. a. Define solar constant. 2  
b. Discuss the factors which cause seasonal change in climate. 6  
c. Describe the earth's thermal balancing process with the help of sketches. 7

**OR**

- IV. a. Define tropical climate. 3  
b. Distinguish between the features of hot dry and warm humid climates. 12

## UNIT – II

- V a. Define metabolism. 3  
b. State the effect of climate on indoor conditions like saturated, still air with above skin temperature. 6  
c. Explain any four subjective variables which influence of thermal comfort of individuals. 6

### OR

VI. Define the following.

1. M R T 2. Specific heat 3. Air to air transmittance 4. Solar heat gain 5. k-value. 15

## . UNIT – III

- VII. a. Explain the effect of green roof in providing passive cooling. 7  
b. Write short notes on water wall and trombe wall 8

### OR

- VIII. a. Explain the problems associated with heating. 6  
b. Describe working of central a/c system. 9

## UNIT – IV

- IX. Illustrate the traditional shelters of hot dry climate and explain their thermal control systems. 15

### OR

- X. Illustrate features of shelters in composite climate 15